Poster ID

3D Object Reconstruction from a Single Image

A proposed approach using image contours and selective completion of hidden regions

Research Goal

The goal is to reconstruct a complete 3D object from one color image. The proposed method encodes image contours and predicts a point cloud, a set of points in 3D space. The current prototype studies contour encoding and point prediction. Completing hidden regions and converting the points into a surface model are broader project goals.

Motivation and Approach

Challenge. A single image shows only the visible surfaces of an object. Reconstructing its complete 3D shape requires estimating depth and filling in hidden regions.

Compact input. Represent object contours with lines and curves, then use these geometric features as inputs to the reconstruction model.

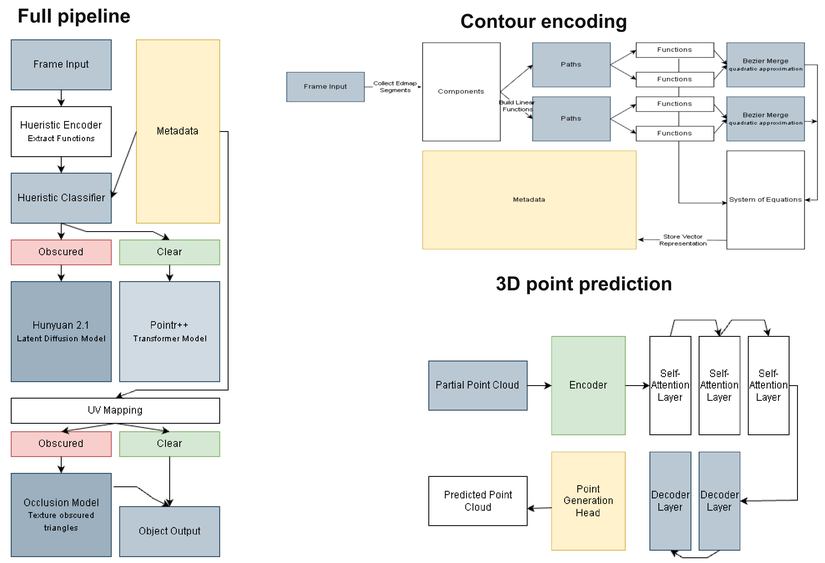
Selective completion. A transformer network predicts 3D points. The proposed design adds a generative model when parts are hidden, with the goal of reducing unnecessary computation.

Final output. The broader goal is to convert the predicted points into a surface mesh and add surface colors. Shape accuracy, surface appearance, and computational savings require separate evaluation.

Research Questions

1. Can image contours provide enough information to reconstruct 3D shape?
2. Can the model complete regions hidden in the input image?
3. Can selective completion reduce computation at matched accuracy?

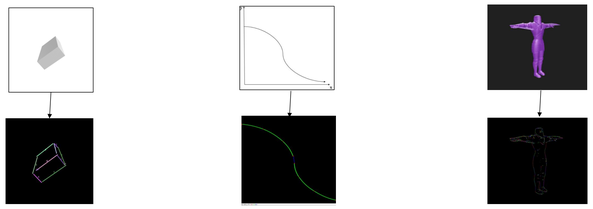
3D Reconstruction Pipeline



Data and Shape Representation

Training data. The project uses Objaverse-XL meshes, which represent object surfaces in 3D [1]. A Blender script renders twelve views per mesh and saves camera and lighting information. These synthetic images provide known 3D geometry for training.

Image contours. A Sobel edge detector extracts contours from each image. The encoder represents straight segments with Ax + By + C = 0 and curved segments with quadratic Bézier curves. Together, these segments form a compact description of image geometry.



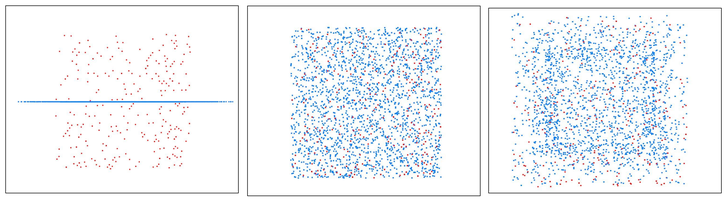
1a. Line-segment representation

1b. Quadratic Bézier curve

1c. Encoded mesh contour, functions separated by color

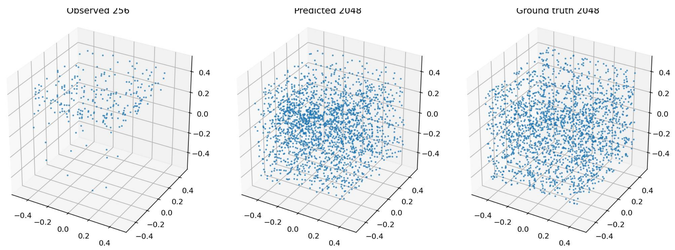
Training coordinates. Using the known mesh and camera pose, the renderer casts rays through sampled contour locations to find points on the 3D surface. The encoder stores these points and groups nearby features for training.

Predicting 3D shape. A transformer network uses the partial geometry to predict a complete point cloud. PoinTr [2] provides related work on completing 3D point clouds with geometric relationships.



2a. Training snapshots at epochs 1, 10, and 30 (left to right). Red: reference points. Blue: predictions.

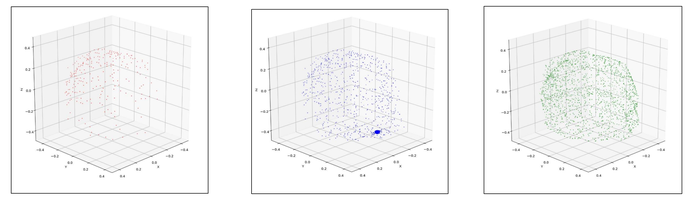
The prototype uses approximately 256 input points and a target point count during training. The cube example below compares partial input, prediction, and ground truth.



2b. Cube reconstruction: partial input, prediction, and ground truth.

Completing Missing 3D Shape

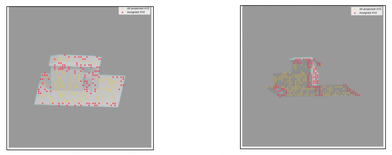
Shape refinement. Repeated network passes refine the predicted 3D points and fill missing regions. The proposed diffusion branch would add a generative model for completing hidden parts.



3. Successive refinement examples (left to right). These visualizations illustrate point-cloud completion.

Training groups. The encoder groups similar contours for training. Neighboring contours provide context because similar 2D outlines can correspond to different 3D shapes.

Using a single image. Training ray casts use a known mesh and camera pose. When given a new image, the pipeline must estimate 3D geometry from the image alone.



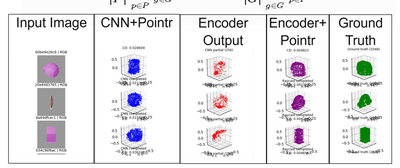
4. Example of encoded coordinates of geometric encoder

Further research. VL-JEPA [5] is a candidate for learning visual features. Its adaptation to 3D completion requires separate evaluation.

Preliminary 3D Reconstruction Results

Chamfer distance compares the predicted 3D points with the reference shape. Lower values mean a closer match under the same normalization.

$$CD(P,G) = \frac{1}{|P|}\sum_{p\in P}\min_{g\in G}\|p-g\|^2 + \frac{1}{|G|}\sum_{g\in G}\min_{p\in P}\|g-p\|^2$$



5. MobileNetV3 was trained on the edge encoder's reference set, containing 1,600 images, with 100 epochs, and a batch size of 4, the Pointr++ model was untouched

MobileNetV3-Small(Smallest Distance)	Heuristic Edge Encoder(Smallest Distance)
0.017	0.035
0.019	0.049
0.036	0.053

References

[1] Deitke et al. Objaverse-XL: A Universe of 10M+ 3D Objects. arXiv:2307.05663, 2023.
[2] Yu et al. PoinTr: Diverse Point Cloud Completion with Geometry-Aware Transformers. ICCV, 2021. arXiv:2108.08839.
[3] Hu and Shugrina. Axolotl3D: a Unified Framework for Faithful 3D Shape Completion. ECCV, 2026.
[4] SAM 3D Team. SAM 3D: 3Dfy Anything in Images. arXiv:2511.16624, 2025 (revised 2026).
[5] Chen et al. VL-JEPA: Joint Embedding Predictive Architecture for Vision-language. arXiv:2512.10942, 2025 (revised 2026).
[6] Shimizu. SweepLSD: A One-Pass, O(width)-Memory Line Segment Detector. arXiv:2608.22086, 2026.

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